The identity and projection operators

A simple matrix-element example

In an orthonormal basis,

$$I_{ij} = \langle i|I|j\rangle = \langle i \mid j\rangle = \delta_{ij}.$$

The full sum $\sum_{i=1}^{n}|i\rangle\langle i|$ equals the identity. What happens if only the first $m < n$ terms are retained?

Projection onto a subspace

Expand an arbitrary vector as

$$\begin{aligned}|v\rangle &= \sum_{i=1}^{n}|i\rangle\langle i \mid v\rangle \\ &= |1\rangle\langle 1 \mid v\rangle + \cdots + |m\rangle\langle m \mid v\rangle + |m+1\rangle\langle m+1 \mid v\rangle + \cdots + |n\rangle\langle n \mid v\rangle.\end{aligned}$$

Keeping only $\sum_{i=1}^{m}|i\rangle\langle i \mid v\rangle$ preserves the first m components and discards those along $|m+1\rangle, \ldots, |n\rangle$. This is the orthogonal projection onto $\mathcal{S} = \text{span}\{|1\rangle, \ldots, |m\rangle\}$:

$$\boxed{P = \sum_{i=1}^{m}|i\rangle\langle i|.}$$

Writing $v_i = \langle i \mid v\rangle$ gives

$$P|v\rangle = \sum_{i=1}^{m} v_i|i\rangle, \qquad \langle v|P = \sum_{i=1}^{m} v_i^*\langle i|.$$

Clarification. The second expression is a bra expansion. The handwritten line places a ket at its end; the bra above restores the correct type.

Projection onto one basis vector

For $P_i = |i\rangle\langle i|$,

$$\begin{aligned}P_iP_j &= |i\rangle\langle i \mid j\rangle\langle j| = \delta_{ij}|i\rangle\langle j| = \delta_{ij}P_i, \\ (P_i)_{k\ell} &= \langle k|P_i|\ell\rangle = \langle k \mid i\rangle\langle i \mid \ell\rangle = \delta_{ki}\delta_{i\ell}.\end{aligned}$$

Thus the only nonzero matrix element of P_i is the unit entry in row i, column i.

The adjoint of an operator

For a scalar multiple, $|u\rangle = \alpha|v\rangle$ implies $\langle u| = \alpha^*\langle v|$. For an operator, define its adjoint by

$$|u\rangle = \Omega|v\rangle \quad \Longrightarrow \quad \langle u| = \langle v|\Omega^\dagger.$$

$\Omega^\dagger$ is also called the *Hermitian conjugate* of Ω.

Matrix elements of the adjoint

Let $|k_i\rangle = \Omega|i\rangle$, so $\langle k_i| = \langle i|\Omega^\dagger$. Then

$$\langle i|\Omega^\dagger|j\rangle = \langle k_i \mid j\rangle = \langle j \mid k_i\rangle^* = \langle j|\Omega|i\rangle^*.$$

Consequently,

$$(\Omega^\dagger)_{ij} = \Omega^*_{ji}, \qquad [\Omega^\dagger] = [\Omega]^{*T}.$$

Taking the adjoint therefore conjugates every matrix element and interchanges rows and columns.

The adjoint reverses a product

$$\boxed{(\Omega\Lambda)^\dagger = \Lambda^\dagger\Omega^\dagger.}$$

To see this, let $|w\rangle = \Lambda|v\rangle$ and $|u\rangle = \Omega|w\rangle = \Omega\Lambda|v\rangle$. Taking adjoints gives

$$\langle u| = \langle w|\Omega^\dagger = \langle v|\Lambda^\dagger\Omega^\dagger.$$

Example: conjugating a vector equation

If

$$\alpha_1|v_1\rangle = \alpha_2|v_2\rangle + \alpha_3|v_3\rangle\langle v_4 \mid v_5\rangle + \alpha_4\Omega\Lambda|v_6\rangle,$$

then

$$\begin{aligned}\alpha_1^*\langle v_1| &= \alpha_2^*\langle v_2| + \alpha_3^*\langle v_4 \mid v_5\rangle^*\langle v_3| + \alpha_4^*\langle v_6|\Lambda^\dagger\Omega^\dagger \\ &= \alpha_2^*\langle v_2| + \alpha_3^*\langle v_5 \mid v_4\rangle\langle v_3| + \alpha_4^*\langle v_6|\Lambda^\dagger\Omega^\dagger.\end{aligned}$$

Classes of operators, determinant, and trace

A matrix is *symmetric* if $A = A^T$. An operator is *Hermitian* if $\Omega^\dagger = \Omega$, and *anti-Hermitian* if $\Omega^\dagger = -\Omega$.

Analogy with complex numbers

Property	Complex number	Operator
Conjugation	$\alpha \mapsto \alpha^*$	$\Omega \mapsto \Omega^\dagger$
Real / Hermitian	$\alpha^* = \alpha$	$\Omega^\dagger = \Omega$
Imaginary / anti-Hermitian	$\alpha^* = -\alpha$	$\Omega^\dagger = -\Omega$
Unit modulus / unitary	$\alpha\alpha^* = 1$	$UU^\dagger = U^\dagger U = I$

Every complex number splits into a real part and a purely imaginary part:

$$\alpha = \underbrace{\frac{\alpha + \alpha^*}{2}}_{\text{real}} + \underbrace{\frac{\alpha - \alpha^*}{2}}_{\text{purely imaginary}} .$$

Similarly, every operator admits the decomposition

$$\Omega = \underbrace{\frac{\Omega + \Omega^\dagger}{2}}_{\text{Hermitian}} + \underbrace{\frac{\Omega - \Omega^\dagger}{2}}_{\text{anti-Hermitian}} .$$

A *unitary* operator satisfies

$$UU^\dagger = U^\dagger U = I.$$

Basis-independent quantities

In a finite-dimensional space, the determinant and trace are

$$\det \Omega = \det(\Omega_{ij}), \qquad \operatorname{Tr} \Omega = \sum_i \Omega_{ii}.$$

Although the matrix entries depend on the basis, these two quantities do not. The trace is invariant under cyclic permutations:

$$\operatorname{Tr}(\Omega\Lambda) = \operatorname{Tr}(\Lambda\Omega),$$

$$\operatorname{Tr}(\Omega\Lambda\Theta) = \operatorname{Tr}(\Theta\Omega\Lambda) = \operatorname{Tr}(\Lambda\Theta\Omega).$$

The eigenvalue problem

In general, $\Omega|v\rangle = |v'\rangle$ changes the direction of a vector. An *eigenvector* is a nonzero vector whose direction is unchanged:

$$\Omega|v\rangle = \omega|v\rangle, \qquad |v\rangle \neq 0.$$

ω is its eigenvalue. For example, $I|v\rangle = |v\rangle$ for every vector, so every nonzero vector is an eigenvector of I with eigenvalue 1. If $\Omega|v\rangle = \omega|v\rangle$, linearity gives

$$\Omega(\alpha|v\rangle) = \omega(\alpha|v\rangle)$$

for every nonzero scalar α.

The characteristic equation

The eigenvalue equation is equivalent to

$$(\Omega - \omega I)|v\rangle = 0.$$

If $\Omega - \omega I$ had an inverse, multiplication by that inverse would give $|v\rangle = 0$, which is excluded. Thus, in finite dimensions,

$$\boxed{\det(\Omega - \omega I) = 0.}$$

To write this in a basis, insert the identity:

$$\begin{aligned} 0 &= \langle i|\Omega - \omega I|v\rangle \\ &= \sum_j \langle i|\Omega - \omega I|j\rangle\langle j \mid v\rangle \\ &= \sum_j (\Omega_{ij} - \omega\delta_{ij})v_j, \qquad v_j = \langle j \mid v\rangle. \end{aligned}$$

The determinant condition becomes

$$\begin{vmatrix} \Omega_{11} - \omega & \Omega_{12} & \cdots & \Omega_{1n} \\ \Omega_{21} & \Omega_{22} - \omega & \cdots & \Omega_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \Omega_{n1} & \Omega_{n2} & \cdots & \Omega_{nn} - \omega \end{vmatrix} = \sum_{m=0}^{n} c_m \omega^m = 0.$$

This polynomial has degree n. Its roots are the eigenvalues, which are independent of the basis used to calculate the determinant.

Finding and normalizing eigenvectors

For a known eigenvalue ω_k, solve the homogeneous linear system

$$\begin{bmatrix} \Omega_{11}-\omega_k & \Omega_{12} & \cdots & \Omega_{1n} \\ \Omega_{21} & \Omega_{22}-\omega_k & \cdots & \Omega_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \Omega_{n1} & \Omega_{n2} & \cdots & \Omega_{nn}-\omega_k \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = 0.$$

For a one-dimensional eigenspace, this fixes the component ratios; some components may be zero. The notes suggest checking the $n = 2$ and $n = 3$ cases directly.

Normalization convention

The solutions $\{v_i\}$ and $\{\alpha v_i\}$ describe proportional eigenvectors, $|v\rangle$ and $\alpha|v\rangle$. Their lengths can differ, so we conventionally choose

$$\langle v \mid v \rangle = 1.$$

Starting with any nonzero eigenvector, the normalized vector is

$$\boxed{|v\rangle_{\text{norm}} = \frac{|v\rangle}{\sqrt{\langle v \mid v \rangle}}.}$$

Normalization leaves an overall phase undetermined.

Repeated eigenvalues

If several eigenvalues coincide,

$$\omega_{k_1} = \omega_{k_2} = \cdots = \omega_{k_m} = \omega_k,$$

the eigenvalue alone need not fix a unique direction: different linearly independent vectors can belong to the same eigenspace. For a Hermitian operator, an eigenvalue of multiplicity m has m linearly independent eigenvectors.

Clarification. The source states this conclusion for a repeated characteristic root without qualifying the operator. In general, algebraic multiplicity need not equal eigenspace dimension. For example, $\left[\begin{smallmatrix} \omega & 1 \\ 0 & \omega \end{smallmatrix}\right]$ has a double root but only one independent eigenvector. The equality used in the following notes holds for Hermitian operators.

Hermitian operators: eigenvalues and orthogonality

Degeneracy

A repeated root of the characteristic equation $\det(\Omega - \omega I) = 0$ has an *algebraic multiplicity*. A double or triple root has multiplicity 2 or 3. In the Hermitian case, this equals the dimension of the corresponding eigenspace, called its degeneracy.

If $|v_1\rangle, \ldots, |v_m\rangle$ are independent eigenvectors with the same eigenvalue ω, then any nonzero combination satisfies

$$\Omega \sum_{a=1}^{m} a_a |v_a\rangle = \omega \sum_{a=1}^{m} a_a |v_a\rangle.$$

Thus every nonzero vector in their span is an eigenvector with eigenvalue ω.

Theorem 10

Every finite-dimensional Hermitian operator has real eigenvalues and admits a complete orthonormal set of eigenvectors.

To prove reality and orthogonality, suppose

$$\Omega|\omega_i\rangle = \omega_i|\omega_i\rangle, \qquad \Omega|\omega_j\rangle = \omega_j|\omega_j\rangle.$$

These equations also allow $i = j$, or $i \neq j$ with $\omega_i = \omega_j$. Taking matrix elements gives

$$\langle\omega_j|\Omega|\omega_i\rangle = \omega_i\langle\omega_j \mid \omega_i\rangle,$$
$$\langle\omega_i|\Omega|\omega_j\rangle = \omega_j\langle\omega_i \mid \omega_j\rangle.$$

Conjugating the second equation and using $\Omega^\dagger = \Omega$ yields

$$\langle\omega_j|\Omega|\omega_i\rangle = \omega_j^*\langle\omega_j \mid \omega_i\rangle.$$

Subtracting the two expressions,

$$\boxed{(\omega_i - \omega_j^*)\langle\omega_j \mid \omega_i\rangle = 0.}$$

The consequences for equal and unequal eigenvalues follow on note 12.

Degenerate eigenspaces and diagonalization

Consequences of Hermiticity

Setting $i = j$ in $(\omega_i - \omega_j^*)\langle\omega_j \mid \omega_i\rangle = 0$ and using $\langle\omega_i \mid \omega_i\rangle > 0$ shows that

$$\omega_i = \omega_i^* : \quad \text{all eigenvalues are real.}$$

For $i \neq j$, the relation becomes

$$(\omega_i - \omega_j)\langle\omega_j \mid \omega_i\rangle = 0.$$

If $\omega_i \neq \omega_j$, the eigenvectors are orthogonal. If $\omega_i = \omega_j$, this equation does not impose orthogonality.

Choosing an orthonormal basis within a degeneracy

Suppose ω is m-fold degenerate. Begin with m linearly independent eigenvectors in that eigenspace. Gram–Schmidt orthonormalization replaces them by linear combinations that are orthonormal. Since these combinations remain in the same eigenspace, they are still eigenvectors with eigenvalue ω:

$$\underbrace{m \text{ independent eigenvectors}}_{\text{same eigenvalue}} \xrightarrow{\text{Gram–Schmidt}} \underbrace{|\omega, 1\rangle, \ldots, |\omega, m\rangle}_{m \text{ orthonormal eigenvectors}} .$$

The second label distinguishes states within the degeneracy. For example, a threefold degeneracy has states $|\omega, 1\rangle, |\omega, 2\rangle, |\omega, 3\rangle$.

Matrix in an eigenvector basis

With an orthonormal eigenbasis,

$$\begin{aligned}\langle\omega', k'|\Omega|\omega, k\rangle &= \omega\langle\omega', k' \mid \omega, k\rangle \\ &= \omega\, \delta_{\omega'\omega}\delta_{k'k}.\end{aligned}$$

Different eigenvalues enforce the first delta; orthonormalization within each eigenspace supplies the second. The matrix is therefore diagonal:

$$[\Omega]_{\text{eigenbasis}} = \begin{bmatrix} \omega_1 & 0 & \cdots & 0 \\ 0 & \omega_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \omega_n \end{bmatrix}.$$

For this reason, solving the eigenvalue problem is often called *diagonalization*.

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Unitary transformation to an eigenvector basis

Let $\{|i\rangle\}$ be an orthonormal basis and let $\{|\omega_i\rangle\}$ be an orthonormal eigenbasis of a Hermitian operator Ω. Different labels may have the same eigenvalue. Define U by

$$|\omega_i\rangle = U|i\rangle, \qquad \langle\omega_i| = \langle i|U^\dagger.$$

Orthonormality gives

$$\langle i|U^\dagger U|j\rangle = \langle\omega_i \mid \omega_j\rangle = \delta_{ij} = \langle i|I|j\rangle.$$

Since $\{|i\rangle\}$ is complete, equality of all matrix elements implies $U^\dagger U = I$. Here U is square, so it is unitary and $UU^\dagger = I$ as well.

Columns of the transformation matrix

The ith component of the jth eigenvector is

$$\langle i \mid \omega_j\rangle = \langle i|U|j\rangle = U_{ij}.$$

Thus, in the original $|i\rangle$ basis,

$$[|\omega_j\rangle] = \begin{bmatrix} \langle 1 \mid \omega_j\rangle \\ \langle 2 \mid \omega_j\rangle \\ \vdots \\ \langle n \mid \omega_j\rangle \end{bmatrix}, \qquad [U] = \begin{bmatrix} \langle 1 \mid \omega_1\rangle & \langle 1 \mid \omega_2\rangle & \cdots & \langle 1 \mid \omega_n\rangle \\ \langle 2 \mid \omega_1\rangle & \langle 2 \mid \omega_2\rangle & \cdots & \langle 2 \mid \omega_n\rangle \\ \vdots & \vdots & \ddots & \vdots \\ \langle n \mid \omega_1\rangle & \langle n \mid \omega_2\rangle & \cdots & \langle n \mid \omega_n\rangle \end{bmatrix}.$$

The jth column of $[U]$ is the coordinate column of the jth normalized eigenvector.

Transforming the operator

$$\langle i|U^\dagger \Omega U|j\rangle = \langle\omega_i|\Omega|\omega_j\rangle = \omega_j\delta_{ij} = \omega_i\delta_{ij}.$$

Therefore

$$[U]^\dagger[\Omega][U] = \mathrm{diag}(\omega_1, \omega_2, \ldots, \omega_n).$$

The columns of $[U]$ transform the original matrix into its diagonal eigenvector representation.

Simultaneous diagonalization of two Hermitian operators

Theorem 13

If Ω and Λ are Hermitian and commute,

$$[\Omega, \Lambda] = 0,$$

there exists a complete orthonormal basis of simultaneous eigenvectors:

$$\Omega|\omega_i\rangle = \omega_i|\omega_i\rangle, \qquad \Lambda|\omega_i\rangle = \lambda_i|\omega_i\rangle.$$

Why each eigenspace is preserved

Starting with $\Omega|\omega_i\rangle = \omega_i|\omega_i\rangle$, multiply by Λ and commute the two operators:

$$\Lambda\Omega|\omega_i\rangle = \omega_i\Lambda|\omega_i\rangle,$$
$$\Omega(\Lambda|\omega_i\rangle) = \omega_i(\Lambda|\omega_i\rangle).$$

Thus $\Lambda|\omega_i\rangle$ remains in the ω_i eigenspace (it may be zero).

(i) If ω_i is nondegenerate, that eigenspace is one-dimensional, so $\Lambda|\omega_i\rangle = \lambda_i|\omega_i\rangle$.

(ii) If ω_i is degenerate, Λ can mix vectors inside that eigenspace, but cannot mix it with an eigenspace of different ω. In particular,

$$\omega_i \neq \omega_j \quad \Longrightarrow \quad \langle\omega_j|\Lambda|\omega_i\rangle = 0.$$

Block structure in an eigenbasis of Ω

Group the equal eigenvalues together. The handwritten block sketches can be written as

$$[\Omega] = \begin{bmatrix} \omega_1 I_{m_1} & 0 & \cdots \\ 0 & \omega_2 I_{m_2} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}, \qquad [\Lambda] = \begin{bmatrix} \Lambda_1 & 0 & \cdots \\ 0 & \Lambda_2 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}.$$

Each Λ_a acts only inside the corresponding ω_a eigenspace. The off-block matrix elements vanish.

Diagonalizing within each degenerate eigenspace

The eigenvalue problem for Λ can be solved independently inside every eigenspace of Ω. Since each restricted block Λ_a is Hermitian, choose an orthonormal eigenbasis for that block. This replaces the old basis by linear combinations within each block, with no mixing between distinct ω eigenspaces.

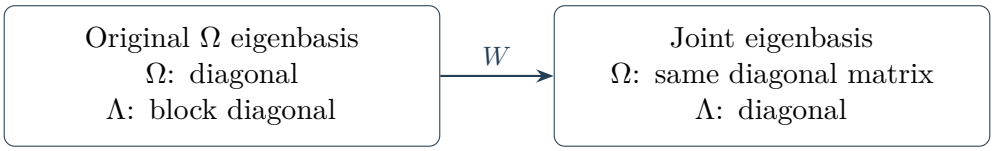
The block transformation

Let W_a diagonalize Λ_a, and combine the transformations as $W = \mathrm{diag}(W_1, W_2, \ldots)$. Then

$$W^\dagger[\Lambda]W = \begin{bmatrix} D_1 & 0 & \cdots \\ 0 & D_2 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}, \qquad D_a = \mathrm{diag}(\lambda_{a1}, \ldots, \lambda_{am_a}).$$

The matrix of Ω is unchanged because every block is a multiple of the identity:

$$W^\dagger[\Omega]W = \mathrm{diag}(\omega_1 I_{m_1}, \omega_2 I_{m_2}, \ldots) = [\Omega].$$



The diagram expresses the source's observation: Ω has the same matrix in the two bases, while the remaining blocks of Λ become diagonal.

Labeling simultaneous eigenvectors

The new vectors are eigenvectors of both operators. A more informative label therefore includes both eigenvalues:

$$\Omega|\omega, \lambda\rangle = \omega|\omega, \lambda\rangle, \qquad \Lambda|\omega, \lambda\rangle = \lambda|\omega, \lambda\rangle.$$

If these two eigenvalues still leave a degeneracy, an additional degeneracy label can be retained. The construction proves the simultaneous-diagonalization statement on note 15a.

Commuting observables and functions of operators

A complete set of commuting observables

When a collection of mutually commuting Hermitian operators resolves all remaining degeneracies, its simultaneous eigenvalues label the joint eigenvectors unambiguously:

$$\Omega, \Lambda, \Gamma, \ldots \quad \longleftrightarrow \quad |\omega, \lambda, \gamma, \ldots\rangle.$$

In the finite-dimensional setting, commuting operators can be added to refine any remaining degenerate subspaces.

Power series

Define Ω^n as the product of n copies of Ω, with $\Omega^0 = I$. If $f(x) = \sum_{n=0}^{\infty} a_n x^n$ converges in the required domain, define

$$f(\Omega) = \sum_{n=0}^{\infty} a_n \Omega^n.$$

The most frequently used example is

$$\mathrm{e}^{\Omega} = \sum_{n=0}^{\infty} \frac{\Omega^n}{n!} = I + \Omega + \frac{\Omega^2}{2!} + \frac{\Omega^3}{3!} + \cdots .$$

If $\Omega|\omega\rangle = \omega|\omega\rangle$, then

$$\mathrm{e}^{\lambda\Omega}|\omega\rangle = \mathrm{e}^{\lambda\omega}|\omega\rangle.$$

For a Hermitian Ω with an orthonormal eigenbasis,

$$\langle\omega'|\mathrm{e}^{\lambda\Omega}|\omega\rangle = \delta_{\omega'\omega}\mathrm{e}^{\lambda\omega},$$

with additional Kronecker deltas for any degeneracy labels.

Differentiation and an operator equation

For λ-independent Ω,

$$\frac{\mathrm{d}}{\mathrm{d}\lambda}\mathrm{e}^{\lambda\Omega} = \Omega\mathrm{e}^{\lambda\Omega} = \mathrm{e}^{\lambda\Omega}\Omega, \qquad \left[\Omega, \mathrm{e}^{\lambda\Omega}\right] = 0.$$

Consequently,

$$\frac{\mathrm{d}\Theta}{\mathrm{d}\lambda} = \Omega\Theta \quad \Longrightarrow \quad \Theta(\lambda) = \mathrm{e}^{\lambda\Omega}\Theta(0).$$

Clarification. For an operator-valued initial constant, its position matters. The exponential belongs on the left for this differential equation. A scalar constant can be placed on either side.

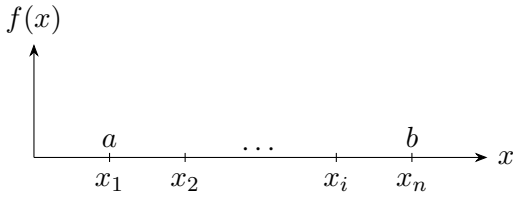
In general $\mathrm{e}^{\Omega}\mathrm{e}^{\Theta} \neq \mathrm{e}^{\Omega+\Theta}$, but

$$[\Omega, \Theta] = 0 \quad \Longrightarrow \quad \mathrm{e}^{\Omega}\mathrm{e}^{\Theta} = \mathrm{e}^{\Omega+\Theta}.$$

From sampled functions to a continuum inner product

Consider a function $f(x)$ on $a \le x \le b$, sampled at n points $x_1, \dots, x_n$. Treat its samples as a coordinate column:

$$[|f_n\rangle] = \begin{bmatrix} f(x_1) \\ f(x_2) \\ \vdots \\ f(x_n) \end{bmatrix}.$$



Introduce an orthonormal discrete coordinate basis $|x_i\rangle$ with

$$\langle x_i \mid x_j \rangle = \delta_{ij}, \qquad \sum_{i=1}^{n} |x_i\rangle\langle x_i| = I.$$

For example, the coordinate column of $|x_\ell\rangle$ has a 1 in position ℓ and zeros elsewhere:

$$[|x_\ell\rangle] = \begin{bmatrix} \langle x_1 \mid x_\ell \rangle \\ \vdots \\ \langle x_\ell \mid x_\ell \rangle \\ \vdots \\ \langle x_n \mid x_\ell \rangle \end{bmatrix} = \begin{bmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{bmatrix}.$$

The corresponding expansions and inner product are

$$\begin{aligned} |g_n\rangle &= \sum_i |x_i\rangle\langle x_i \mid g_n\rangle = \sum_i g(x_i)|x_i\rangle, \\ \langle f_n| &= \sum_i f(x_i)^* \langle x_i|, \\ \langle f_n \mid g_n\rangle &= \sum_i f(x_i)^* g(x_i). \end{aligned}$$

As the sampling grows denser, the unweighted sum need not have a finite limit. Introduce the Riemann-sum weight $\Delta_n = (b-a)/n$:

$$\langle f_n \mid g_n\rangle_\Delta = \sum_i f(x_i)^* g(x_i) \Delta_n.$$

For suitable functions the continuum limit is

$$\boxed{\langle f \mid g\rangle = \lim_{n\to\infty} \sum_i f(x_i)^* g(x_i) \Delta_n = \int_a^b f(x)^* g(x)\, \mathrm{d}x.}$$

Clarification. The weighted inner product is a new convention; the associated coordinate basis must be rescaled consistently in the continuum limit. Note 18 gives its Dirac-delta normalization.

Continuum normalization and the Dirac delta

The norm of a function is the special case

$$\langle f \mid f \rangle = \int_a^b |f(x)|^2 \, \mathrm{d}x.$$

Discrete orthogonality, $\langle x_i \mid x_j \rangle = 0$ for $i \neq j$, becomes orthogonality for distinct continuum labels. The completeness relation becomes

$$\int_a^b |x'\rangle\langle x'| \, \mathrm{d}x' = I.$$

Taking its matrix element between $\langle x|$ and an arbitrary state $|f\rangle$ gives

$$\int_a^b \langle x \mid x' \rangle \langle x' \mid f \rangle \, \mathrm{d}x' = \langle x \mid f \rangle,$$
$$\int_a^b \langle x \mid x' \rangle f(x') \, \mathrm{d}x' = f(x).$$

Since this kernel reproduces every suitable test function,

$$\boxed{\langle x \mid x' \rangle = \delta(x - x').}$$

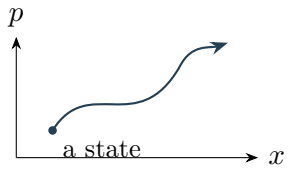
The Dirac delta vanishes away from $x = x'$ in the distributional sense, and its integral action supplies the required normalization. The continuum basis vectors are generalized eigenvectors, rather than ordinary unit-norm Hilbert-space vectors.

Chapter 4: the postulates of quantum mechanics

The four postulates concern (I) the state, (II) observables, (III) measurement, and (IV) time evolution.

I. State

Classical mechanics. A particle is specified by its position and momentum (x, p), a point in phase space. A succession of states traces a phase-space trajectory.



Quantum mechanics. A pure state is represented by a nonzero vector $|\psi\rangle$ in the physical Hilbert space, conventionally normalized to unit norm. Proportional vectors represent the same ray.

II. Observable

A classical observable is a real-valued function $\omega(x, p)$. Quantum observables are represented by Hermitian operators. In particular,

$$X|x\rangle = x|x\rangle, \qquad P|p\rangle = p|p\rangle,$$

$$[X, P] = \mathrm{i}\hbar I, \qquad [X_i, P_j] = \mathrm{i}\hbar\delta_{ij} I, \qquad [X_i, X_j] = [P_i, P_j] = 0.$$

The source denotes the classical-to-quantum correspondence by $\Omega(X, P) = \omega(x \to X, p \to P)$.

Clarification. For products of noncommuting variables, operator ordering must also be specified; simple substitution alone does not settle it.

III. Measurement

In a classical state (x_0, p_0), measuring ω yields $\omega(x_0, p_0)$ and, ideally, leaves the state unchanged. In a quantum state $|\psi\rangle$, measuring Ω yields one of its eigenvalues ω. For a normalized state and a nondegenerate discrete eigenvalue,

$$P(\omega) = |\langle\omega \mid \psi\rangle|^2, \qquad |\psi\rangle \xrightarrow{\text{measurement}} |\omega\rangle.$$

The probability is the squared magnitude of the component along the corresponding eigenvector. Degenerate outcomes are treated by projection onto the entire eigenspace (note 21).

Time evolution and position and energy measurements

IV. Time evolution

Classical evolution follows Hamilton's equations,

$$\dot{x} = \frac{\partial \mathcal{H}}{\partial p}, \qquad \dot{p} = -\frac{\partial \mathcal{H}}{\partial x}.$$

Quantum evolution follows the Schrödinger equation,

$$\boxed{\mathrm{i}\hbar \frac{\mathrm{d}}{\mathrm{d}t}|\psi(t)\rangle = H|\psi(t)\rangle.}$$

Here $H(X, P)$ is the Hamiltonian operator corresponding to $\mathcal{H}(x, p)$, with an appropriate ordering of noncommuting factors when necessary.

States and normalization

Use $\langle \psi \mid \psi \rangle = 1$ for a given physical state. If a state is constructed as $|\psi\rangle = \Omega|\phi\rangle$, its norm need not be 1; when it is nonzero, replace it by $|\psi\rangle/\sqrt{\langle \psi \mid \psi \rangle}$. The position basis is complete and delta normalized:

$$\langle x' \mid x \rangle = \delta(x - x'), \qquad \int |x\rangle\langle x| \, \mathrm{d}x = I.$$

In three spatial dimensions, write $|x, y, z\rangle = |\mathbf{r}\rangle$. Then

$$\langle \mathbf{r}' \mid \mathbf{r} \rangle = \delta^{(3)}(\mathbf{r} - \mathbf{r}'), \qquad \int |\mathbf{r}\rangle\langle \mathbf{r}| \, \mathrm{d}^3 r = I,$$

$$|\psi\rangle = \int |\mathbf{r}\rangle\langle \mathbf{r} \mid \psi\rangle \, \mathrm{d}^3 r = \int \psi(\mathbf{r})|\mathbf{r}\rangle \, \mathrm{d}^3 r.$$

Position measurement

Measuring (X, Y, Z) gives a position in the volume element $\mathrm{d}^3 r$ with probability

$$\mathrm{d}P = |\langle \mathbf{r} \mid \psi \rangle|^2 \, \mathrm{d}^3 r = |\psi(\mathbf{r})|^2 \, \mathrm{d}^3 r.$$

An ideal exact-position outcome is represented formally by $|\mathbf{r}\rangle$. Measurement is a state-selection rule, not simply the action $\mathbf{R}|\psi\rangle$ of the position operator.

Energy measurement

For a discrete nondegenerate spectrum,

$$|\psi\rangle = \sum_n |E_n\rangle\langle E_n \mid \psi\rangle, \qquad \phi_n(\mathbf{r}) = \langle \mathbf{r} \mid E_n \rangle.$$

An energy measurement gives E_n with probability

$$\begin{aligned} P(E_n) &= |\langle E_n \mid \psi \rangle|^2 \\ &= \left| \int \langle E_n \mid \mathbf{r}\rangle\langle \mathbf{r} \mid \psi\rangle \, \mathrm{d}^3 r \right|^2 = \left| \int \phi_n(\mathbf{r})^* \psi(\mathbf{r}) \, \mathrm{d}^3 r \right|^2. \end{aligned}$$